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Course:

Diploma in Artificial Intelligence and Machine Learning

Report Name:

Spam Mail Prediction Using Machine Learning

Academic Year: 2026

Acknowledgement

I sincerely thank my project guide, faculty members, and internship mentors for their valuable guidance and continuous support during the completion of this project. I also thank my institute for providing the necessary resources and learning environment. Finally, I express my gratitude to my family and friends for their constant encouragement.

Abstract

Spam emails are one of the major challenges in digital communication. They contain unwanted advertisements, phishing links, and malicious content that can compromise user security. This project presents a Spam Mail Prediction System using Machine Learning and Natural Language Processing (NLP). The email text is converted into numerical features using TF-IDF Vectorization, and Logistic Regression is used to classify emails as Spam or Ham. The developed model achieves high prediction accuracy and demonstrates how machine learning can improve email security through automatic filtering.

Introduction

Email is one of the most widely used communication methods. However, spam emails have become increasingly common, affecting both individuals and organizations. Traditional filtering methods often fail to detect new spam patterns. Machine Learning provides an intelligent approach by learning from existing email data and automatically identifying spam messages.

This project focuses on building a spam detection system using text classification techniques.

Problem Statement

Manual identification of spam emails is inefficient and unreliable. Large volumes of emails make it difficult to separate important messages from spam. Therefore, there is a need for an automated system that accurately classifies emails as Spam or Ham using machine learning techniques.

Objectives

1. Build a spam mail prediction system.
2. Apply Natural Language Processing techniques.
3. Convert text into numerical features using TF-IDF.
4. Train a Logistic Regression classifier.
5. Evaluate prediction performance.
6. Predict whether a new email is Spam or Ham.

Dataset Description

The dataset contains email messages labeled into two categories.

Input Feature:

- ➢ Message

Output Feature:

- Category
- Spam
- Ham

The dataset includes hundreds of email samples used for model training and testing.

Technologies Used

Programming Language:

- Python

Libraries:

- NumPy
- Pandas
- Matplotlib
- Seaborn
- Scikit-learn
- Pickle

Machine Learning Algorithm:

- Logistic Regression

Natural Language Processing:

- TF-IDF Vectorizer

Development Environment:

- Google Colab
- Jupyter Notebook

Methodology

The project follows these steps:

1. Import required libraries.
2. Load the email dataset.
3. Handle missing values.
4. Convert labels into numerical values.
5. Split the dataset into training and testing sets.
6. Convert email text using TF-IDF Vectorization.
7. Train the Logistic Regression model.
8. Evaluate model accuracy.
9. Predict new email messages.

Data Preprocessing

The dataset was cleaned by replacing missing values with blank strings. Email categories were converted into numerical labels. The text data was transformed into numerical vectors using TF-IDF Vectorization, enabling the machine learning model to process textual information effectively.

Model Building

Logistic Regression was selected because it performs efficiently for binary classification tasks. The dataset was divided into 80% training data and 20% testing data. The TF-IDF Vectorizer converted textual data into numerical features before model training.

Advantages

1. Automatically filters unwanted emails.
2. Improves email security.
3. High prediction accuracy.
4. Fast classification process.
5. Reduces manual effort.

Limitations

1. Performance depends on dataset quality.
2. Cannot detect every new spam pattern.
3. Requires retraining with updated datasets.
4. Limited to binary classification.

Future Scope

1. Deploy as a web application.
2. Build a browser extension.
3. Develop a mobile spam detection application.
4. Train deep learning models for higher accuracy.
5. Support multiple languages.

Conclusion

The Spam Mail Prediction System was successfully developed using Natural Language Processing and Logistic Regression. TF-IDF Vectorization transformed email text into meaningful numerical features, allowing the model to classify emails accurately. The project demonstrates the practical application of Machine Learning in cybersecurity and email filtering, providing an efficient solution for detecting spam messages.

References

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3. NumPy Documentation
4. Matplotlib Documentation
5. Seaborn Documentation
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